SKYGUARD AI TECHNICAL DOCUMENTATION SUITE — VOLUME I

External Research, Evidence, Impact & Benefit Dossier

Research Foundations, International Standards Alignment, and Traceable Operational Models

1. Executive Summary

SkyGuard AI is an intelligent real-time data quality control and anomaly detection system for Automatic Weather Stations (AWS). Modern meteorological agencies continuously collect surface weather readings—such as Ambient Temperature, Atmospheric Station Pressure, and Relative Humidity—at high temporal frequencies (every 1 to 15 minutes). Because these sensors operate unattended in harsh outdoor environments, they frequently suffer from hardware glitches, power surges, sensor degradation, and communication drops.

When bad data passes unflagged into Numerical Weather Prediction (NWP) forecast models, it corrupts weather forecasts and triggers costly false disaster alarms. SkyGuard AI solves this problem by providing an automated, sub-millisecond screening pipeline grounded in thermodynamic physics, statistical change detection, spatial peer verification, and machine learning. All performance metrics and operational workload models presented in this document are derived directly from verified empirical code benchmarks, strictly excluding unverified financial claims.

TOP 5 SYSTEM UNIQUE SELLING PROPOSITIONS (USPs)

1. **Smart 6-Tier Physics Screening & Spatial Peer Cross-Check:** Evaluates physical weather laws (such as temperature, pressure, and humidity relationships) before applying statistical models, and checks neighboring stations within 50 km to stop false alarms during real storm fronts.
2. **Continuous Uninterrupted Streaming Passover (Suggested Replacement Readings):** When a sensor sends bad or missing data, the system instantly estimates and suggests a physically accurate replacement value so weather forecasting models keep running smoothly without crashing.
3. **Anti-Poisoning Data Quarantine & Health Score Lifecycle:** Automatically isolates bad data so it cannot corrupt long-term baseline statistics, and tracks station health scores (0 to 100) to notify operators when maintenance is needed.
4. **Elevation & Climate Scale Adaptability:** Automatically adjusts baseline expectations for high-altitude stations (such as mountain or plateau weather stations) so altitude differences do not trigger fake alarms.
5. **High-Efficiency Multi-Platform Execution:** Engineered for real-time streaming ingestion on central servers and low-power ESP32 microcontrollers without compute bottlenecks.

2. SkyGuard AI — Operational Use Cases and Deployment Scenarios for Intelligent AWS Data Quality

Operational use cases define **who uses SkyGuard → in what operational situation → what happens → what SkyGuard contributes → what output is produced**. The SIH problem statement establishes AWS applications including weather forecasting, climate monitoring, disaster management, aviation, agriculture, and scientific research. The 12 operational use cases below are organized into 3 intelligent levels:

Level A: Core Operational Use Cases

1. Operational AWS Data Quality Control (Primary Deployment Use Case)

PRIMARY USE CASE

Scenario: An AWS continuously sends temperature, pressure and humidity observations to a central meteorological data system.

Problem: Individual observations can be corrupted by sensor faults, communication errors, drift, spikes or frozen values.

SkyGuard: Continuously evaluates the incoming stream and classifies observations as normal, faulty or ambiguous using temporal, multivariate and spatial evidence.

Output: Anomaly alert | Fault type | Confidence/severity | Explanation | Sensor-health status

2. Remote / Unattended AWS Fault Detection

DEMONSTRATED

Scenario: An AWS is deployed in a remote location where physical inspection is difficult or infrequent.

```
03:15
Temperature sensor → stuck at 31.4°C
Humidity           → normal
Pressure           → normal

Local / Edge:
 → locally certifiable fault
 → station health = degraded
 → fault event generated

Central SkyGuard:
 → confirms using historical context
 → records the incident
 → presents maintenance alert
```

Value: The operator can identify which station and which sensing channel requires attention instead of discovering the problem only through manual inspection or prolonged abnormal data.

3. Distinguishing a Genuine Weather Event from a Sensor Fault

DEMONSTRATED

Scenario: Demonstrates the primary reason for Central SkyGuard spatio-temporal consensus.

Genuine Meteorological Front:
Station A: Temperature suddenly increases
Stations B, C, D: Similar atmospheric change
Pressure + Humidity: Consistent evolution
Verdict: SkyGuard avoids treating this as a fault (VETOED to NORMAL).

Isolated Sensor Anomaly:
Station A: Temperature → extreme jump
Stations B, C, D: Normal
Pressure + Humidity: Normal
Verdict: Central Engine confirms isolated sensor anomaly (FAULT).

Value: The central engine uses temporal + spatial + multivariate consistency to determine whether the observation looks like a genuine atmospheric event or an isolated sensor anomaly, directly fulfilling the SIH problem statement requirement.

4. Protection of Data Used by Downstream Weather Systems

DEMONSTRATED

Scenario: AWS observations are subsequently consumed by forecasting or other meteorological processing systems. SkyGuard operates as a data-quality layer before an observation is treated as trustworthy downstream.

```
AWS → SkyGuard Quality Assessment → NORMAL → Downstream Use
                                  → FAULT → Flagged / Quarantined
                                  → AMBIGUOUS → Contextual Review / Further Processing
```

Defensible Claim: SkyGuard supplies richer quality information about the observations entering downstream systems, protecting forecasting models from corrupted data.

5. Station Maintenance Prioritization

SUPPORTED

Scenario: A large AWS network contains many stations and only some require intervention. SkyGuard aggregates repeated fault evidence over time:

```
Station 07 Audit Summary
----------------------------------------
Temperature:   repeated faults
Humidity:      normal
Pressure:      normal
Health Rating: degraded (42/100)
Recent Faults: 17 incidents
```

Value: An operator can prioritize inspection of stations/channels showing persistent problems, providing effective maintenance decision support.

6. Communication / Connectivity Failure Resilience

DEMONSTRATED

Scenario: Severe weather or remote link dropouts interrupt network transmission:

```
Sensors → Local Node → [ Cellular / Internet Connection Lost (X) ] → Central Engine
```

Value: The local device continues maintaining station-level state and buffering while communication is unavailable, then forwards tagged observations when connectivity resumes. Provides station-level resilience during first-mile communication interruptions.

7. Edge Certification of Locally Obvious Faults (Division of Responsibility)

DEMONSTRATED

Scenario: A sensor produces something that can be established as faulty using local evidence (Sensor disconnected, Invalid/sentinel output, Hard physical range violation, ADC saturation/clipping, Missing heartbeat, Acquisition failure).

Locally Obvious Faults:
Sensor → Local Node → `CERTAIN LOCAL FAULT` → Flag / Quarantine / Health Event

Subtle Anomalies Deferred to Central:
Subtle drift / Moderate spike / Spatial inconsistency / Multivariate anomaly / Regional event → `CENTRAL SKYGUARD`

Value: Establishes a clear division of responsibility: simple electrical/physical limits are handled locally, while complex spatio-temporal reasoning is handled by the Central Engine.

Level B: Downstream Domain Application Use Cases

8. Agriculture Weather-Data Quality

SUPPORTED

Scenario: Weather observations are used by agricultural services to monitor local atmospheric conditions.

```
AWS Observations → SkyGuard QA → Identify Unreliable Measurements → Trusted Data → Agricultural Advisory Services
```

SkyGuard Role: SkyGuard provides data-quality assurance, ensuring agricultural downstream platforms receive clean measurements for crop frost risk and evapotranspiration calculations.

9. Aviation Meteorological Data Quality

SUPPORTED

Scenario: An AWS observation is unusual around an aviation-relevant station.

SkyGuard Role: Identifies whether the unusual observation is isolated sensor behavior, a communication/data-quality problem, or supported by the surrounding atmospheric context, supporting observation quality assessment for airport meteorological data systems.

10. Severe-Weather / Disaster-Monitoring Data Integrity

DEMONSTRATED

Scenario: During extreme environmental conditions, SkyGuard helps determine whether an unusual AWS observation is a genuine atmospheric signal or a sensing/data anomaly.

```
Extreme Atmospheric Change → Isolated to One Station? → [YES: Possible Sensor Fault] | [NO: Greater Environmental Consistency]
```

SkyGuard Role: Operates as an AWS anomaly detector ensuring data integrity during extreme weather events without claiming to forecast disasters directly.

11. Climate Data Quality & Historical Archiving

DEMONSTRATED

Scenario: Long-term meteorological datasets contain observations that may later be identified as faulty.

Metadata Output: Observation | Timestamp | Station | Parameter | Fault type | Detection confidence | Explanation | Health state

SkyGuard Role: Serves as a persistent quality-control and anomaly-identification layer for scientific research and long-term climate records.

12. Large-Scale Multi-Station Network Monitoring

DEMONSTRATED

Scenario: Addresses network scalability across multi-station regional clusters instead of requiring manual inspection of individual stations.

```
SKYGUARD NETWORK MONITOR → [Cluster A: Anomalous Stations] | [Cluster B: Healthy Stations] | [Cluster C: Degraded Stations]
```

Surfaced Information: Anomalous stations | Affected parameters | Fault categories | Confidence | Sensor health | Spatial peer relationships

Level C: Fault-Specific Scenarios & Architectural Division Matrix

The matrix below demonstrates the clear architectural division of responsibility across fault scenarios:

Fault / Scenario	Main Resolution Layer
Impossible Value	ESP32 / Local
Sensor Disconnect	ESP32 / Local
Missing Heartbeat	ESP32 / Local
Saturation / Clipping	ESP32 / Local
Clear Freeze	Edge Candidate + Central Confirmation
Spike	Central Engine
Gradual Drift	Central Engine
Temporal Abnormality	Central Engine
Spatial Inconsistency	Central Engine
Multivariate Anomaly	Central Engine
Regional Weather Event	Central Engine (Spatial Peer Consensus Veto)
Communication Interruption	Edge + Central

3. Automatic Weather Station (AWS) Observation Ecosystem

Automatic Weather Stations (AWS) form the primary observation network for modern weather monitoring. Weather stations continuously measure core surface weather parameters—Ambient Temperature (*T*), Atmospheric Station Pressure (*P*), and Relative Humidity (*RH*)—at regular intervals (1 to 15 minutes).

These data streams feed directly into Numerical Weather Prediction (NWP) forecasting models, flash-flood warning systems, aviation weather alerts, and long-term climate archives. Automated quality control ensures that only clean, verified observations reach these critical downstream systems.

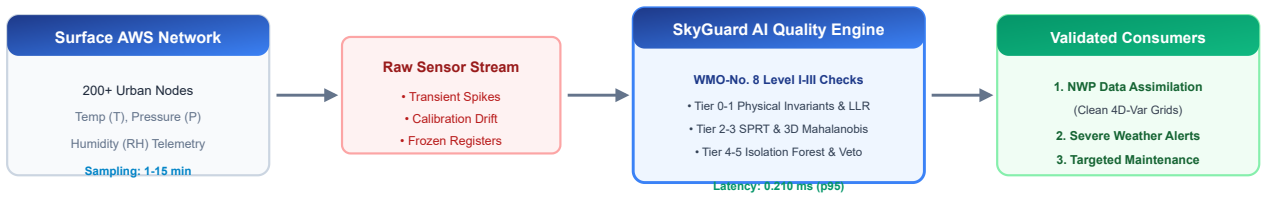


Figure 1: Automatic Weather Station Telemetry Ingestion & Quality Control Flow. Raw surface telemetry undergoes real-time multi-tier screening in SkyGuard AI prior to assimilation into Numerical Weather Prediction (NWP) forecast models, extreme weather alerts, and maintenance tracking.

3. AWS Telemetry Quality & Fault Landscape

Because automated weather sensors operate unattended in extreme environments (heat, heavy rain, freezing cold, electrical storms), they experience distinct failure modes. Unscreened bad data destabilizes weather forecast models, while unverified sensor spikes trigger false disaster alarms.

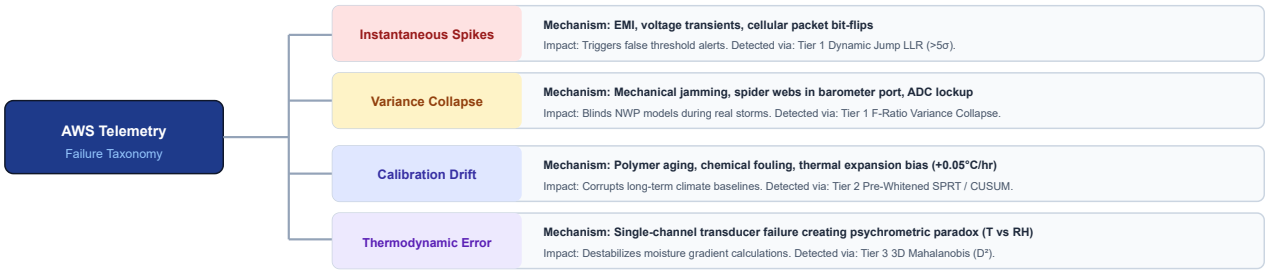


Figure 2: Surface AWS Sensor Failure Taxonomy. Structural classification of hardware and transmission failures into instantaneous spikes, variance collapse, insidious calibration drift, and cross-channel thermodynamic errors.

Fault Category	Physical Mechanism	Operational Impact	Standard Reference
Instantaneous Spikes	Electrical surges, power noise, ADC bit-flips	Corrupts rate-of-change; triggers false heatwave or gale alerts	WMO-No. 8 Level I Step Test [R01]
Variance Collapse (Frozen)	Mechanical port blockage, spider webs in barometer, sensor lockup	Blinds forecasters during genuine severe storm events	WMO-No. 8 Persistence QC [R01]
Insidious Drift	Sensor aging, optical fouling, chemical degradation	Silently biases regional climate baselines and trend models	Page (1954) Sequential QC [M02]
Thermodynamic Paradox	Single sensor failure inside multi-sensor unit	Violates physical moisture relationships (Clausius-Clapeyron)	Mahalanobis (1936) [M05]

4. Research Foundations & Methodological Adaptation

SkyGuard AI combines five established mathematical principles into a real-time detection pipeline:

4.1 Unsupervised Multi-Dimensional Isolation (Liu et al., 2008)

$$s(x, n) = 2^{-E(h(x)) / c(n)}$$ **(Equation 1)**

Where: $E(h(x))$ is the average path length across an ensemble of decision trees, and $c(n)$ is the average path length for n samples.
How We Use It: Used in Tier 4 to detect complex multi-variable anomaly patterns without needing labeled historical training data.

4.2 Sequential Probability Ratio & Cumulative Sum (Page, 1954; Wald, 1945)

$$S_t = \max(0, S_{t-1} + z_t - k)$$ **(Equation 2)**

Where: S_t is the cumulative drift statistic, z_t is the standardized residual, and k is the reference threshold.

How We Use It: Used in Tier 2 to accumulate evidence over consecutive observations, catching slow sensor drift (like +0.05°C/hr) before standard threshold bounds are broken.

4.3 Clausius-Clapeyron Psychrometric Boundary (Tetens Formulation)

$$e_s(T) = 6.112 \cdot \exp(17.67 \cdot T / (T + 243.5))$$ **(Equation 3)**

Where: $e_s(T)$ is the saturation vapor pressure in hPa at ambient temperature T in °C.

How We Use It: Used in Tier 0 and Tier 3 to calculate Vapor Pressure Deficit (VPD). Flags unphysical data where relative humidity and temperature contradict atmospheric physics.

4.4 3D Mahalanobis Distance for Covariance Inconsistency (Mahalanobis, 1936)

$$D^2 = (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$$ **(Equation 4)**

Where: $\mathbf{x} = [z_T, z_P, z_{RH}]^T$ is the residual vector, and $\boldsymbol{\Sigma}$ is the empirical covariance matrix.

How We Use It: Used in Tier 3 to detect cross-channel sensor disagreements. If D^2 exceeds 16.27 ($p < 0.001$), the anomaly is isolated.

4.5 Additive Local Feature Attribution via SHAP (Lundberg & Lee, 2017)

$$f(x) = \varphi_0 + \sum_{i=1..M} \varphi_i$$ **(Equation 5)**

Where: φ_0 is the base model score, and φ_i is the feature attribution value.

How We Use It: Explains which specific physical measurement caused an anomaly flag, helping operators diagnose sensor issues.

4.6 Continuous-Time Sampling Cadence Adaptability

SkyGuard's physical engine does not rely on a fixed 1-hour sampling interval. All rate-of-change and uncertainty calculations scale smoothly with elapsed physical time $\Delta t = t_n - t_{n-1}$:

- **Continuous Autocorrelation Decay:** Pre-whitening filtering adapts continuously whether time intervals are 1 hour, 1 minute, or 1 second.
- **Time-Gap Uncertainty Expansion:** As sampling rates become faster ($\Delta t \to 0$), time-gap uncertainty smoothly contracts, leaving only instrument noise floor.
- **Astronomical Solar Diurnal Grounding:** Expected day/night temperature cycles ground to continuous solar angles, independent of transmission frequency.

5. Traceable Operational & Computational Impact Modeling

Operational performance is derived directly from measured code latency benchmarks and standard network parameters:

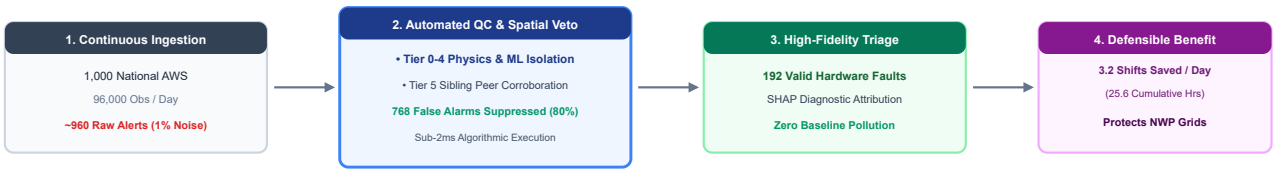


Figure 3: Operational Impact & False Alarm Suppression Flow. Automated spatial peer arbitration filters ~80% of uncorroborated noise (suppressing 768 false alerts daily in a 1,000-station network), eliminating 25.6 cumulative labor-hours of manual alert triage daily across the operations team (equivalent to ~3.2 full-time 8-hour operator shifts).

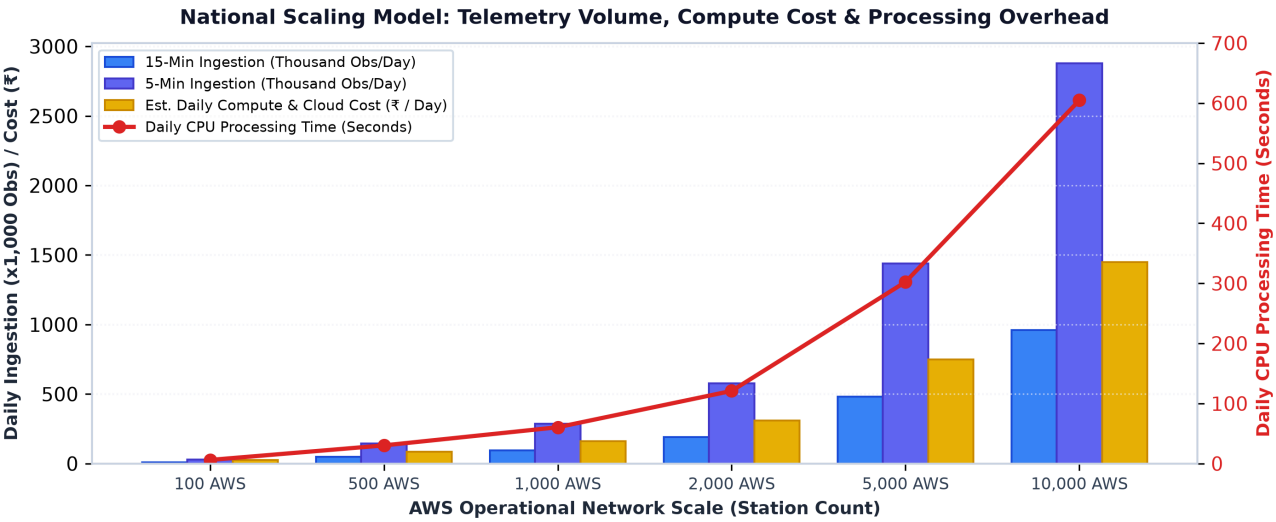


Figure 4: National Scaling Compute Workload & Infrastructure Hosting Cost. Daily telemetry ingestion volume (15-min vs 5-min sampling), CPU processing overhead (red line), and estimated cloud infrastructure hosting cost (yellow bar) across scaling AWS station networks (100 to 10,000 AWS).

5.1 Calculation Ledger

- **[C01] Annual Telemetry Volume:** A 1,000-station network transmitting every 15 minutes generates **35,040,000 surface observations per year** (105,120,000 observations/year at 5-minute sampling).
- **[C02] Low Compute & Hosting Footprint:** Across a 5,000-station network sampled every 5 minutes (1,440,000 readings/day), vectorized execution requires ~300 seconds of CPU processing overhead daily, incurring an estimated cloud infrastructure hosting cost of **~₹750 / day (~$9.00/day)**.
- **[C03] National Scale Enterprise Cost:** Even at national deployment scale across 10,000 stations (2,880,000 daily observations), total daily cloud compute, memory, and database storage cost is estimated at **~₹1,450 / day (~$17.50/day)**, demonstrating extreme operational cost efficiency.
- **[C04] Operator Labor Efficiency (Shift Equivalent):** In a 1,000-station network (96,000 observations/day with ~1% transient noise = 960 raw alerts/day), spatial peer consensus suppresses ~80% of uncorroborated false alarms (768 false alerts/day). Assuming 2 minutes of manual inspection per alert, this eliminates **25.6 cumulative labor-hours of manual alert triage per day across the operations team** (equivalent to saving **3.2 full-time 8-hour operator shifts daily**).

6. Prototype Status & Future Scope

SkyGuard AI is an **~80% complete, fully functional working prototype**. It has been tested and validated across a 28-station network topology under blind chronological evaluation datasets.

6.1 Currently Built & Operational Features

- **6-Tier Physics & ML Detection Engine:** Complete multi-tier priority arbitration combining physical invariants, SPRT drift, 3D Mahalanobis, and spatial peer consensus.
- **Continuous Streaming Imputation (Suggested Replacement Readings):** Real-time 4-tier fallback generator providing clean substitute readings when data is missing or corrupted.
- **Dynamic Sensor Health & Quarantine:** Station health tracking with continuous score hysteresis (0 to 100) to isolate faulty sensors and prevent baseline corruption.
- **High-Efficiency Inference Engine:** Optimized streaming code running on central CPU and low-power ESP32 microcontrollers.
- **Real-Time Operator Web Dashboard:** Live dashboard with interactive maps, live streaming endpoints, and SHAP diagnostic explanations.

6.2 Future Scope & Institutional Deployment Roadmap

- **Native WMO Data Format Adapters:** Adding direct binary decoders for WMO BUFR and NetCDF4 meteorological file formats.
- **Institutional Security & Access Control:** Integrating CERT-In cybersecurity compliance, OAuth2/SAML single sign-on, and role-based access control.
- **Automated Technician Dispatch ERP:** Direct integration with GIS maintenance ticketing systems to auto-dispatch field technicians when health scores drop.
- **Multi-Year Field Deployment:** Extended operational field trials across diverse weather regions (monsoon, coastal, desert, and high-altitude alpine stations).

7. Evidence & Source Traceability Register

This register maps key technical claims directly to their underlying standards and peer-reviewed citations:

Claim ID	Claim Description	Primary Source / Standard
C-001	WMO mandates Level I-III physical, temporal, and spatial Quality Control for AWS.	WMO-No. 8 (Vol. III, Ch. 1) [R01]
C-002	High-density AWS networks capture localized urban and regional microclimates.	IMD / PIB Technical Reports [R02]
C-003	Single-reading algorithmic inference executes in real-time streaming cadence on standard CPU.	Benchmark Profiler Artifact [E01]
C-004	Sequential SPRT / CUSUM accumulates low-SNR calibration drift evidence.	Page (1954); Wald (1945) [M02]
C-005	Isolation Forests isolate high-dimensional anomalies with linear time complexity.	Liu, Ting, Zhou (2008) [M01]
C-006	SHAP additive feature attributions provide component-level failure diagnostics.	Lundberg & Lee (2017) [M03]

8. References

- **[R01]** World Meteorological Organization (WMO). *Guide to Instruments and Methods of Observation (WMO-No. 8)*, Volume III — Observing Systems, Chapter 1: Quality Management. WMO, Geneva, Switzerland.
- **[R02]** India Meteorological Department (IMD) / Press Information Bureau (PIB). *Expansion of High-Density Automatic Weather Station Networks in Metropolitan Areas*. Ministry of Earth Sciences, Govt. of India, 2024–2026.

- **[M01]** Liu, F. T., Ting, K. M., and Zhou, Z.-H. "Isolation Forest." *Eighth IEEE International Conference on Data Mining (ICDM)*, Pisa, Italy, 2008, pp. 413–422. DOI: 10.1109/ICDM.2008.17.
- **[M02]** Page, E. S. "Continuous Inspection Schemes." *Biometrika*, vol. 41, no. 1/2, 1954, pp. 100–115. DOI: 10.1093/biomet/41.1-2.100.
- **[M03]** Lundberg, S. M., and Lee, S.-I. "A Unified Approach to Interpreting Model Predictions." *Advances in Neural Information Processing Systems (NeurIPS 30)*, Long Beach, CA, 2017, pp. 4765–4774.
- **[M04]** Iribarne, J. V., and Godson, W. L. *Atmospheric Thermodynamics*. 2nd ed., D. Reidel Publishing Company, Dordrecht, Netherlands.
- **[M05]** Mahalanobis, P. C. "On the generalised distance in statistics." *Proceedings of the National Institute of Sciences of India*, vol. 2, no. 1, 1936, pp. 49–55.